ARTIFICIAL INTELLIGENCE GROUP • UNIVERSITY OF HAGEN

Argumentation-based Causal Reasoning

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POWERED BY **TweetyProject** reasoning libraries & solvers

WEB TOOL • OPEN SOURCE

From causal models to argumentation-based explanations.

The Causal Knowledge Base Editor is a visual interface for constructing causal knowledge bases and **reasoning** over them through **structured argumentation**, making explicit the competing causal inferences.

AVAILABLE AT

causal-knowledge-base-editor.aig.fernuni-hagen.de

WORKFLOW

1 Build a causal model

Define unobservable and explainable causal variables, connect them through Boolean causal relationships.

2 Induce an AF

Based on the causal model and observations, derive causal arguments and connect them through undercut attacks.

3 Reason & explain

Compute stable extensions, evaluate conclusions, and generate dependency- and sequence-based explanations.

FORMAL BACKGROUND

1 Causal knowledge bases

DEFINITION

Let U, V partition the atoms into **background** and **explainable** atoms. A **Boolean causal model** is a triple (U, V, K); a **causal knowledge base** is $\Delta = (K, A)$.

$$K = \{v \leftrightarrow \varphi_v \mid v \in V\}$$ **Boolean structural equations** *(one equation per explainable atom)*

$$A \subseteq \mathrm{Lit}(U), \quad \forall u \in U : \ A \cap \{u, \neg u\} \neq \varnothing$$ **Background assumptions** *(at least one literal per background atom)*

2 Induced argumentation frameworks

DEFINITION

Given $\Delta = (K, A)$, the **induced AF** is $F(\Delta) = (\mathrm{Args}(\Delta), \Rightarrow)$.

An **argument** is a pair $a = (\Phi, \psi) \in \mathrm{Args}(\Delta)$:

$$\Phi \subseteq A, \quad K \cup \Phi \nvdash \bot, \quad K \cup \Phi \vdash \psi, \quad \forall \Psi \subsetneq \Phi : \ K \cup \Psi \nvdash \psi$$ **Causal Argument** *(ψ negates a premise of a')*

For arguments $a = (\Phi, \psi)$ and $a' = (\Phi', \psi')$, an **attack** is defined by:

$$a \Rightarrow a' \iff \exists \phi' \in \Phi' : \ \phi' \equiv \neg\psi$$ **Undercut attack** *(premises are consistent and minimal for ψ)*

3 Semantics-based conclusions

DEFINITION

Given $F = (\mathrm{Args}, \Rightarrow)$, a **stable extension** is a set $E \subseteq \mathrm{Args}$:

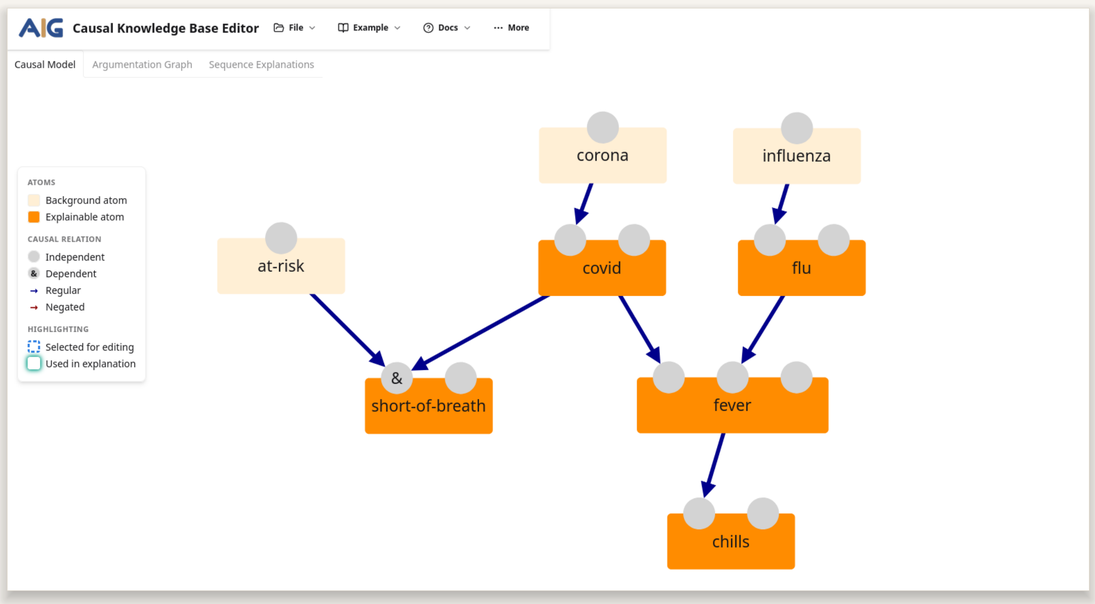
$$\forall a, b \in E : \ a \not\Rightarrow b, \quad \forall a \in \mathrm{Args} \setminus E \ \exists b \in E : \ b \Rightarrow a$$ **Stable extension** *(conflict-free; attacks all outside arguments)*

For an observation φ and conclusion ψ, **causal entailment** is defined by:

$$\varphi \mid\!\sim_\Delta \psi \iff \forall E \in \mathsf{st}(F(K \cup \{\varphi\}, A)) : \ \exists (\Phi, \psi) \in E$$ **Skeptical acceptance** *(ψ is supported in every stable extension)*

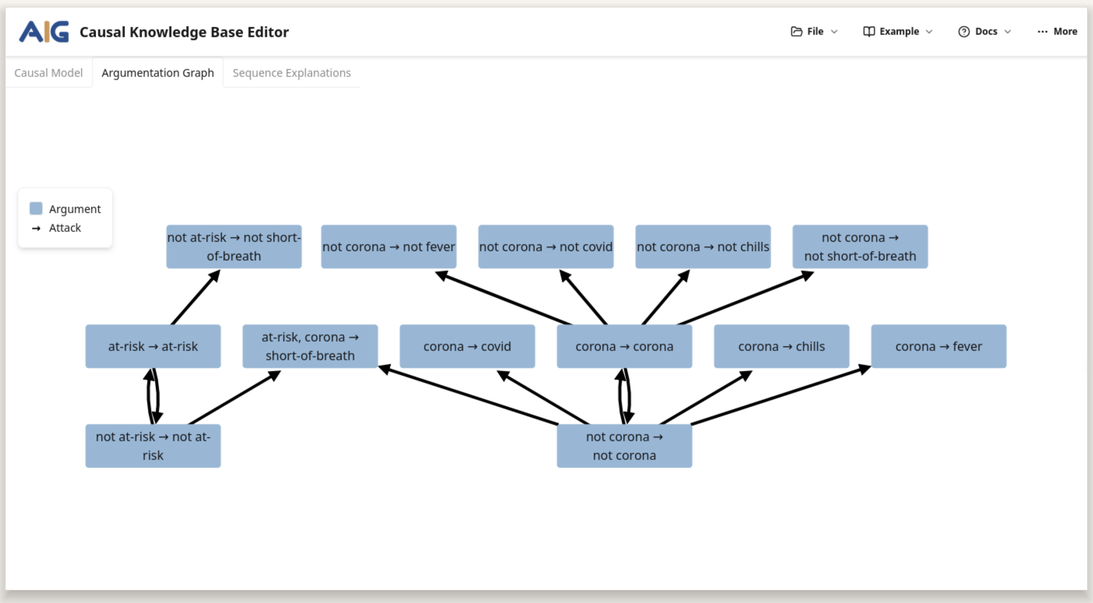
IN ACTION

1 Causal model



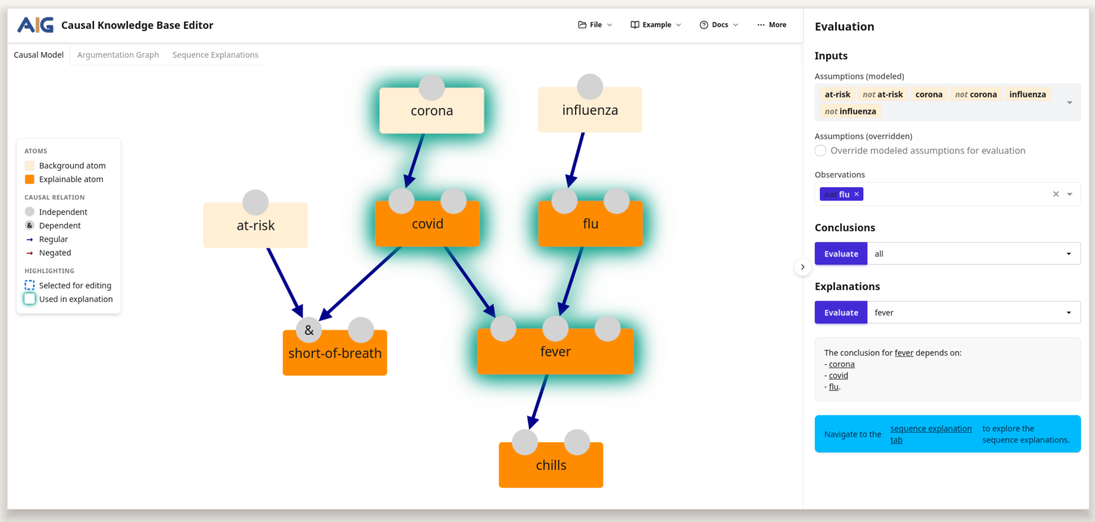
The example: corona causes covid, influenza causes flu, either leads to fever then chills, and covid in an at-risk patient also causes shortness of breath. **Background atoms** (light) are unobservable; **explainable atoms** (orange) are derived. A **&** port is a **conjunctive cause**, otherwise causes are **independent**.

2 Induced argumentation framework



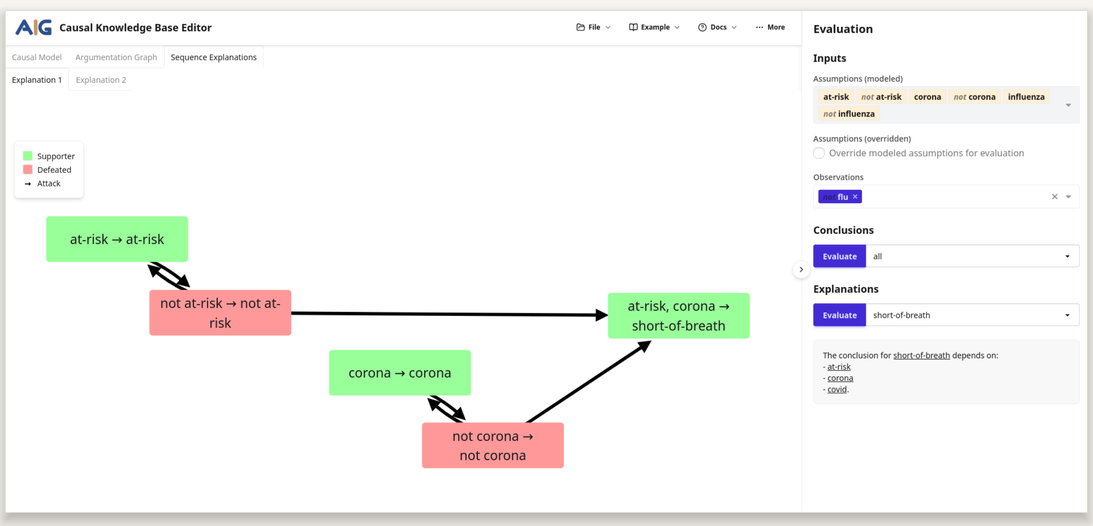
The AF induced by this model given the observation that the patient has **not flu**. Nodes are **causal arguments**, edges **undercut attacks**.

3 Reasoning & explanations



The **evaluation sidebar** lists the **causal conclusions** that follow from the model and the observation **not flu**. Highlighted nodes trace the **causal chain** deriving *fever*.

4 Sequence-based explanations



A **sequence explanation**: a tree-like derivation of *short-of-breath* from the **causal arguments** of the induced AF, again given the observation **not flu**.

OUTLOOK

- **Presentation of Explanations.** Integrate results more tightly with the induced framework, making it easier to inspect how arguments and attacks lead to accepted conclusions.
- **Counterfactual reasoning.** Add counterfactual reasoning via the twin-model method to contrast the actual model with an intervened copy and reveal how changed conditions affect conclusions.

causal-knowledge-base-editor.aig.fernuni-hagen.de • **github.com/aig-hagen/aig-causal-knowledge-base-editor**

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